FLAT FAMILIES LOCALLY DOMINATED BY A QUASI-COMPACT OPEN ARE FPQC COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3235, 3246] and proof(s) [[3248, 3258]]; this spec covers the complete environment instead (statement 3235-3246, proof span(s) [(3248, 3258)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original lemma 03L7: sources/stacks-chow/tags.txt, source lines 4703–4703. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03L8: sources/stacks-chow/tags.txt, source lines 4704–4704. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be a scheme. Let $\{f_i : T_i \to T\}_{i\in I}$ be a family of morphisms of schemes with target T. Assume that*

1. *each f_i is flat, and*
2. *there exists an fpqc covering $\{g_j : S_j \to T\}_{j\in J}$ such that each $\{S_j \times_T T_i \to S_j\}_{i\in I}$ is an fpqc covering.*

Then $\{f_i : T_i \to T\}_{i\in I}$ is an fpqc covering of T.

Proof. We will use Lemma 03L7 without further mention. Let $U \subset T$ be an affine open. By (2) we can find quasi-compact opens $V_j \subset S_j$ for $j \in J$, almost all empty, such that $U = \bigcup g_j(V_j)$. Then for each j we can choose quasi-compact opens $W_{ij} \subset S_j \times_T T_i$ for $i \in I$, almost all empty, with $V_j = \bigcup_i \mathrm{pr}_1(W_{ij})$. Thus $\{S_j \times_T T_i \to T\}$ is an fpqc covering. Since this covering refines $\{f_i : T_i \to T\}$ we conclude by Lemma 03L8. □

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